

Digital Retail Banking Platform (DRBP)

BUSINESS ANALYSIS CASE STUDY

Project Information

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1. Project Overview

Project Name: Digital Retail Banking Platform (DRBP)

Industry: BFSI (Banking, Financial Services, and Insurance)

Client: ABC Bank Ltd.

Project Duration: 10 months

Methodology: Agile (Scrum)

- Sprint Duration: 2 Weeks
- Total Sprints: 20

Role: Senior Business Analyst

Responsibilities:

- Conduct stakeholder interviews
- Gather business requirements
- Facilitate requirement workshops
- Document BRD and FRD
- Write user stories and acceptance criteria
- Create AS-IS and TO-BE process flows
- Prepare wireframe requirements
- Manage the Requirement Traceability Matrix (RTM)
- Support Sprint Planning and Sprint Review

- Collaborate with developers, testers, and Product Owner
- Support User Acceptance Testing (UAT)
- Assist with production rollout

Project Summary:

The Digital Retail Banking Platform is designed to provide customers with secure and convenient banking services through web and mobile applications. The platform enables digital account onboarding, account management, fund transfers, bill payments, deposits, card services, and loan applications. The project aims to enhance customer experience, reduce dependency on branch visits, automate manual processes, and improve operational efficiency while ensuring compliance with banking regulations and security standards.

2. Business Problem

ABC Bank Ltd. has experienced a significant increase in customers using digital channels, but its existing online banking platform is outdated, slow, and lacks key self-service features. Customers must visit branches for account opening, KYC verification, fixed deposits, and loan applications, resulting in long wait times and high operational costs.

The bank also faces challenges with:

- Poor customer experience due to multiple disconnected systems.
- High customer support calls for routine banking requests.
- Delays in digital onboarding because of manual KYC verification.
- Low adoption of digital services.
- Security concerns related to authentication and fraud detection.

To remain competitive with digital-first banks and fintech companies, the bank plans to implement a modern Digital Retail Banking Platform.

3. Business Objectives

Primary Objective

Develop a secure, user-friendly, and scalable Digital Retail Banking Platform that enables customers to perform everyday banking services online, reducing dependency on branch visits while improving customer satisfaction and operational efficiency.

Detailed Business Objectives

1. Increase digital adoption by enabling customers to access banking services through web and mobile applications.
2. Reduce branch visits by **40%** within **12 months** by offering end-to-end digital self-service capabilities.
3. Decrease customer onboarding time from **2 days to less than 15 minutes** using digital eKYC and online account opening.
4. Enable secure and real-time transactions, including fund transfers, bill payments, deposits, and card management.
5. Improve customer satisfaction by providing an intuitive user experience and 24×7 banking availability.
6. Reduce operational costs by automating manual banking processes and minimizing paper-based workflows.
7. Strengthen security through Multi-Factor Authentication (MFA), encryption, fraud detection, and real-time transaction monitoring.
8. Ensure compliance with banking regulations, KYC, AML, and data privacy requirements.
9. Provide personalized banking services such as transaction insights, spending summaries, and tailored product recommendations.
10. Improve employee productivity by reducing manual processing and allowing branch staff to focus on higher-value customer interactions.

4. Scope

The project aims to design and implement a Digital Retail Banking Platform that enables retail customers to securely access banking services through web and mobile applications. The platform will provide digital self-service capabilities for account management, payments, transfers, deposits, loans, and customer support while ensuring security, regulatory compliance, and a seamless user experience.

In Scope

- Digital account opening via eKYC (document upload + video KYC)
- Secure login with multi-factor authentication (OTP/biometric)
- Consolidated post-login dashboard with account, card, and alert summary
- Account summary — balances, statements, transaction history
- Fund transfer (NEFT, RTGS, IMPS) with beneficiary management
- Bill payment with biller registration and payment history

- Fixed deposit booking, tenure/interest calculation, premature closure
- Credit card activation, limit view, statement access, block/replace requests
- Personal loan eligibility check, application, and status tracking
- Transactional and service notifications (SMS/email/in-app)
- Customer support via ticketing and chatbot/FAQ assistance
- Admin portal for internal user, application, and platform management

Out of Scope

- Wealth management and investment product services (e.g., mutual funds, insurance cross-sell) .
- Corporate and business banking services (platform is retail-customer only) .
- Replacement of existing IVR and call-centre systems (platform will integrate with, not replace, these during transition)
- Self-service credit limit increases above the bank's pre-approved threshold (requires separate approval workflow per BR-25)
- Merchant/POS-side payment processing
- Third-party account aggregation (e.g., viewing accounts held at other banks)

5. Assumptions & Constraints

Assumptions

- Customers have a smartphone or computer with internet access and a valid government ID for eKYC.
- Core banking system (CBS) exposes APIs for real-time balance, transaction, and account data.
- NPCI/UPI and NEFT/RTGS gateways are accessible through existing bank infrastructure.
- Relationship managers and support staff will be trained before platform rollout.
- Existing customer data can be migrated from legacy systems without loss.

Constraints

- Must comply with RBI's Master Direction on Digital Payment Security Controls and KYC norms.
- MFA must add no more than one extra step to the login flow, per the bank's UX guidelines.
- Legacy data migration is dependent on the core banking vendor's export timelines.

- Platform must integrate with existing IVR and call-center systems during transition, not replace them outright.
- Delivery timeline is fixed by the bank's board-approved digital transformation roadmap.

6. Risk

Risk	Impact	Likelihood	Mitigation
CBS API integration delays	Delays core banking data availability, pushing back launch	Medium	Early integration testing with core banking vendor; phased API rollout
Customer resistance to eKYC/digital onboarding	Slower adoption than projected, undermining onboarding targets	Medium	In-app guidance, branch-assisted digital onboarding option during transition
MFA-related login friction	Increased drop-off or support calls if OTP delivery is unreliable	Medium	Multiple MFA channels (SMS + biometric); fallback OTP resend flow
Regulatory non-compliance (RBI KYC/security norms)	Legal penalties, mandated platform changes post-launch	Low	Compliance officer sign-off at each phase; periodic regulatory review
Data migration errors from legacy systems	Incorrect balances/history shown to customers, trust erosion	Medium	Data validation and reconciliation before cutover; parallel run period
Fraud attempts during transition period	Financial loss, reputational damage	Medium	Real-time transaction monitoring, fraud team review of flagged logins
Scope creep from stakeholder requests	Timeline and budget overrun	High	Change request process tied to board-approved roadmap; BA-led scope sign-off
Third-party payment gateway downtime	Failed transactions, customer complaints	Low	SLA-backed gateway contracts; fallback to secondary provider

7. Stakeholder Analysis

Stakeholder	Role	Interest/Stake	Influence	Involvement
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Retail Customer	End user	Fast onboarding, secure access, self-service banking	Low (indirect, via feedback/adoption)	End-to-end platform user
Relationship Manager	Sales/service	Digital tools to support onboarding, FD, and loan follow-ups	Medium	Consulted during onboarding and lending flow design
Loan Processing Team	Operations	Efficient, accurate digital loan application intake	Medium	Reviews and processes applications via Admin Portal
Customer Support Executive	Support	Reduced call volume; tools to resolve escalated tickets	Medium	Uses ticketing/chatbot module; provides UAT feedback
Fraud & Risk Team	Risk management	Strong authentication, transaction monitoring	High	Defines MFA and fraud-flagging requirements
Compliance Officer	Regulatory oversight	Adherence to RBI KYC and security norms	High	Sign-off on eKYC and security design
IT/Core Banking Team	Technical delivery	Stable CBS integration, system uptime	High	Owns API integration and platform infrastructure
System Administrator	Platform operations	Manageable user roles, monitoring, configuration	Medium	Primary user of Admin Portal
Bank Leadership/Board	Sponsor	ROI on digital transformation roadmap, customer retention	High	Approves scope, budget, and timeline

Business Analyst	Project delivery	Accurate requirements, stakeholder alignment	Medium	Elicits requirements, manages scope, documents traceability
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8. Requirement Gathering Approach

1. Overview

This document outlines the approach used to elicit, analyze, and validate requirements for the Digital Retail Banking Platform. Given the platform spans 12 modules involving multiple internal teams (compliance, fraud & risk, IT, loan processing) and external-facing customer journeys, a mixed-method elicitation approach was adopted — combining stakeholder interviews, workshops, document analysis, and process observation — to ensure requirements were both business-accurate and technically feasible before being finalized into the Functional and Non-Functional Requirements.

2. Objectives of Requirement Gathering

- Identify pain points in the bank's existing digital channels (net banking, mobile app, IVR) directly from the teams and customers affected by them.
- Capture functional needs for each of the 12 modules from the stakeholders who will use or be impacted by them.
- Surface regulatory and security constraints early, before design work begins.
- Validate assumptions made in the Project Overview and Scope documents against real operational input.

3. Stakeholder Engagement

Stakeholder engagement was structured around the stakeholder groups identified in the Stakeholder Analysis, with the elicitation method matched to each group's role and availability:

Stakeholder	Engagement Method	Purpose
Retail Customers	Surveys + usability interviews	Understand pain points in current onboarding,

		transfers, and support experience
Relationship Managers	Focus group workshop	Capture friction points in FD/loan sales that require branch follow-up
Loan Processing Team	One-on-one interviews	Document current loan intake, document verification, and approval workflow
Customer Support Executives	Shadowing + interviews	Identify recurring, high-volume queries suitable for self-service or chatbot deflection
Fraud & Risk Team	Workshop	Define MFA requirements and fraud-flagging rules
Compliance Officer	Document review + interview	Confirm RBI KYC and security norms that must be built into eKYC and login flows
IT/Core Banking Team	Technical workshop	Assess CBS API availability and integration constraints
Bank Leadership	Kickoff and sign-off meetings	Align on business objectives, scope boundaries, and success criteria

4. Elicitation Techniques Used

- **Stakeholder Interviews** — one-on-one sessions with Loan Processing, Compliance, and IT teams to capture role-specific requirements and constraints.
- **Workshops** — joint sessions with Fraud & Risk and IT teams to define MFA flow and security requirements collaboratively.
- **Document Analysis** — review of existing net banking portal specifications, RBI KYC guidelines, and past customer complaint logs.
- **Process Observation** — shadowing customer support executives handling live calls to identify which queries could be moved to self-service.
- **Surveys** — sent to a sample of existing customers to gauge digital adoption barriers and feature priorities.
- **Prototype Walkthroughs** — low-fidelity wireframes reviewed with Relationship Managers and Support Executives to validate assumptions before finalizing FRs.

5. Requirement Validation

- Draft requirements were reviewed in a **sign-off workshop** with Compliance, Fraud & Risk, and IT stakeholders to confirm feasibility and regulatory alignment.
- Functional requirements were traced back to specific stakeholder inputs to ensure no requirement was added without a documented source.
- Ambiguous or conflicting requirements (e.g., MFA friction vs. security strictness) were resolved through a follow-up workshop before being finalized in the FR/NFR document.

6. Tools Used

- Requirement documentation: structured user story templates (as used in the User Stories document)
- Traceability: requirements-to-stakeholder mapping maintained in a traceability matrix
- Prototyping: wireframes for key journeys (onboarding, fund transfer, loan application)

9. User Roles & Permissions

Role	Access Scope
Retail Customer	Full self-service access to their own account: Registration & eKYC, Login, Dashboard, Account Summary, Fund Transfer, Bill Payment, Fixed Deposit, Credit Card Management, Personal Loan Application, Notifications, Customer Support
Relationship Manager	Read-only view of assigned customers' account and product summary; cannot initiate transactions on a customer's behalf
Loan Processing Team	Full access to Personal Loan Application queue (view, review, approve/reject); read access to applicant eligibility and documents via Admin Portal
Customer Support Executive	Access to Customer Support ticketing queue and chatbot escalations; restricted, view-only access to a customer's account summary for the duration of an active ticket
Fraud & Risk Team	Full access to fraud and MFA alert dashboards in Admin Portal; can suspend accounts flagged for suspicious activity; no access to unrelated customer data
Compliance Officer	Full access to Admin Portal audit reports and eKYC review queue; can approve/reject flagged registrations; read-only access to transaction logs for audit purposes

System Administrator	Full administrative access: user role management, platform configuration, monitoring dashboard; no access to individual customer financial data beyond system logs
IT/Core Banking Team	Technical access to system logs, API monitoring, and infrastructure health; no access to customer-identifiable financial data
Bank Leadership/Board	No direct system access; receives reported metrics (KPI dashboards, audit summaries) outside the platform
Business Analyst	Read access to all modules for requirements validation and UAT; no transactional or administrative privileges in production

Access Rules

- BR-36: Role assignment shall be mandatory for every internal user before Admin Portal access is granted (ties to BR-33).
- BR-37: A single internal user shall not hold more than one role simultaneously unless explicitly approved by the System Administrator.
- BR-38: Retail Customers shall have zero visibility into other customers' data, regardless of shared household or joint account status, unless explicitly configured as a joint account.
- BR-39: Role changes shall take effect immediately and shall be logged for audit (ties to BR-34).
- BR-40: Fraud & Risk and Compliance roles shall have override authority to suspend a customer account; no other internal role shall have this permission.

10. Business Rules

Module 1: Customer Registration & eKYC

- BR-01: Customer must provide a valid PAN and Aadhaar number; the system shall reject registration if either fails UIDAI/NSDL validation.
- BR-02: Customer must be at least 18 years old to open an individual savings account.
- BR-03: Uploaded documents must be JPEG, PNG, or PDF, and must not exceed 5MB per file.
- BR-04: A mismatch between the uploaded document and the ID database shall route the application to manual compliance review — it shall not be auto-approved.

- BR-05: One Aadhaar/PAN combination shall be linked to only one active retail account.

Module 2: Login with MFA

- BR-06: Every login shall require a second authentication factor (OTP or biometric); single-factor login is not permitted.
- BR-07: An OTP shall expire 5 minutes after issuance and cannot be reused.
- BR-08: The account shall lock after 3 consecutive failed MFA attempts and require identity re-verification to unlock.
- BR-09: A new device or location login shall trigger an additional verification step.

Module 3: Dashboard

- BR-10: The dashboard shall display only accounts and cards linked to the logged-in customer's profile.
- BR-11: Balance figures shown on the dashboard must reflect real-time data from the core banking system, not cached values older than 5 minutes.

Module 4: Account Summary

- BR-12: Statement generation shall be limited to a maximum range of 3 years from the current date.
- BR-13: Statements shall be available for download only after successful MFA-verified login.

Module 5: Fund Transfer

- BR-14: Daily transfer limits shall apply per customer, per transfer mode (NEFT/RTGS/IMPS), as configured by the bank.
- BR-15: A transfer request exceeding the available balance shall be rejected before submission to the payment gateway.
- BR-16: A new beneficiary shall be subject to a cooling period (e.g., 30 minutes to 24 hours) before the first transfer can be made to them.
- BR-17: RTGS transfers shall only be permitted for amounts of ₹2,00,000 or above; amounts below this threshold shall default to NEFT or IMPS.

Module 6: Bill Payment

- BR-18: A biller must be registered with the Bharat Bill Payment System (BBPS) to be eligible for listing.
- BR-19: A bill payment shall not be processed twice for the same biller reference number within 24 hours, to prevent duplicate payment.

Module 7: Fixed Deposit

- BR-20: Minimum FD booking amount shall be ₹1,000; minimum tenure shall be 7 days.
- BR-21: Premature closure shall attract a penalty as per the bank's prevailing FD closure policy, deducted from the payout amount.
- BR-22: Interest earned on FDs shall be subject to TDS as per applicable income tax rules once the threshold is crossed.

Module 8: Credit Card Management

- BR-23: A credit card shall only be activated after successful MFA verification tied to the registered mobile number.
- BR-24: A blocked card shall not be reactivated; only a replacement card issuance is permitted.
- BR-25: Credit limit changes shall require a separate approval workflow and cannot be self-serviced above the bank's pre-approved threshold.

Module 9: Personal Loan Application

- BR-26: Loan eligibility shall be computed based on declared income, existing obligations, and credit bureau (CIBIL) score.
- BR-27: An applicant with an active loan default on record shall be auto-rejected at the eligibility check stage.
- BR-28: Loan disbursement shall only occur after all required documents are verified and approved by the processing team.

Module 10: Notifications

- BR-29: A transactional alert shall be sent within 1 minute of the triggering event, regardless of channel.

- BR-30: A customer shall be able to opt out of promotional notifications but not transactional/security alerts.

Module 11: Customer Support

- BR-31: A support ticket unresolved after 48 hours shall be automatically escalated to a senior support agent.
- BR-32: Chatbot interactions unresolved after 2 failed attempts shall be automatically routed to a live agent.

Module 12: Admin Portal

- BR-33: Internal user access shall be governed by role-based permissions; no internal user shall have access beyond their assigned role.
- BR-34: All administrative actions (user creation, role changes, application approvals) shall be logged with timestamp and user ID for audit purposes.
- BR-35: Audit logs shall be retained for a minimum of 7 years, in line with RBI record-retention norms.

11. Functional Requirements

ID	Module	Requirement
FR-01	Customer Registration & eKYC	System shall allow customers to register using PAN/Aadhaar-based eKYC with document upload and video verification
FR-02	Customer Registration & eKYC	System shall validate uploaded documents and flag mismatches for manual review
FR-03	Login with MFA	System shall authenticate users via credentials followed by OTP or biometric verification
FR-04	Login with MFA	System shall lock the account after 3 consecutive failed login attempts
FR-05	Dashboard	System shall display a consolidated view of account balances, recent transactions, and active alerts on login
FR-06	Account Summary	System shall allow customers to view balances, download statements, and search transaction history by date/type
FR-07	Fund Transfer	System shall support NEFT, RTGS, and IMPS transfers with beneficiary addition and daily limit checks

FR-08	Fund Transfer	System shall allow customers to schedule one-time or recurring transfers
FR-09	Bill Payment	System shall allow biller registration and payment with confirmation and receipt generation
FR-10	Fixed Deposit	System shall allow FD booking with tenure and interest calculation, and support premature closure requests
FR-11	Credit Card Management	System shall allow card activation, limit viewing, statement access, and block/replacement requests
FR-12	Personal Loan Application	System shall allow customers to check loan eligibility, submit applications with documents, and track status
FR-13	Notifications	System shall send transactional and service alerts via SMS, email, and in-app notification
FR-14	Customer Support	System shall allow customers to raise support tickets and access FAQ/chatbot assistance
FR-15	Admin Portal	System shall allow administrators to manage customer records, review applications, and configure platform settings
FR-16	Admin Portal	System shall provide role-based access control for internal users

12. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Dashboard and Account Summary pages shall load within 3 seconds under normal load
NFR-02	Security	All customer data in transit and at rest shall be encrypted (TLS 1.2+, AES-256)
NFR-03	Security	MFA shall be enforced on all customer logins without exception
NFR-04	Availability	Platform shall maintain 99.9% uptime, excluding scheduled maintenance windows
NFR-05	Compliance	System shall comply with RBI's Master Direction on Digital Payment Security Controls and KYC norms
NFR-06	Scalability	System shall support concurrent access by up to 500,000 active users without degradation
NFR-07	Usability	Core transactions (login, transfer, bill payment) shall be completable within 4 screens or fewer
NFR-08	Auditability	All financial transactions and admin actions shall be logged with timestamp and user ID for audit purposes
NFR-09	Compatibility	Platform shall be accessible via major web browsers and on Android/iOS mobile devices

NFR-10	Recoverability	System shall support data backup and recovery with a maximum tolerable downtime of 4 hours
NFR-11	Usability/Security	MFA shall add no more than one additional step to the standard login flow, regardless of the second-factor method chosen (OTP or biometric)

13. Process Flow

Process Name

Customer Onboarding — Branch-Based Account Opening (As-Is)

Objective

To document the current manual, branch-dependent process by which a retail customer opens a new savings account at ABC Bank Ltd. from initial branch visit through account activation, debit card issuance, and internet banking access.

Actors

- **Customer** — individual seeking to open a new account
- **Customer Service Representative (CSR)** — first point of contact at the branch
- **Branch Officer** — verifies and approves the application
- **Operations Team** — performs backend data entry and card/internet banking requests
- **Core Banking System (CBS)** — legacy system of record for account creation

Step-by-Step Current Process (20 Steps)

1. Customer visits the branch during working hours with original ID and address proof.
2. Customer takes a token and waits in the general queue.
3. CSR greets the customer and understands the requirement (new account opening).
4. CSR manually verifies the customer's original identity and address proof documents (physical KYC check).
5. CSR hands over a blank paper account-opening application form to the customer.
6. Customer manually fills out the paper form (personal details, nominee, account type).
7. Customer attaches self-attested photocopies of ID proof, address proof, and a passport-size photograph.
8. CSR reviews the filled form for completeness and accuracy.
9. **Decision Point:** If the form is incomplete or documents are missing, CSR returns it to the customer for correction (loop back to Step 6).

10. If complete, CSR logs the application manually in a physical register and forwards it to the Branch Officer.
11. Branch Officer manually cross-checks the application against the submitted documents.
12. **Decision Point:** Branch Officer approves or rejects the application based on document authenticity and eligibility (e.g., age, existing account checks).
13. If rejected, the customer is informed verbally or via a follow-up call; process ends.
14. If approved, Branch Officer forwards the physical file to the Operations Team.
15. Operations Team manually keys the customer's data into the Core Banking System (CBS).
16. CBS generates a new account number and stores the customer record.
17. Operations Team submits a manual request to the card division for debit card issuance.
18. Debit card is printed centrally and dispatched to the customer's registered address via courier (typically 7–10 business days).
19. Operations Team separately submits a request to activate internet banking, which is provisioned manually within CBS.
20. Customer receives account details via a physical welcome letter and/or SMS notification once the account is active.

Decision Points

- Application form completeness check (Step 9)
- Branch Officer approval/rejection based on document verification (Step 12)

Manual Activities

- Physical KYC document verification
- Paper form filling by customer
- Manual logging in a physical register
- Manual cross-checking of documents by Branch Officer
- Manual data entry into CBS by Operations Team
- Manual, separate requests for debit card and internet banking activation

System Interactions

- CBS: customer record creation, account number generation

- Card printing/dispatch system: triggered manually, not integrated with CBS in real time
- Internet banking provisioning module: activated manually by Operations Team, not automatic on account creation
- SMS/notification system: triggered only after account is fully active

Pain Points and Bottlenecks

- Customer must visit the branch physically and cannot begin the process online.
- Long, unpredictable wait times due to token-based queueing.
- Form errors discovered only after manual review, requiring resubmission and repeat visits.
- No real-time verification of ID/address proof — relies entirely on human judgment.
- Multiple handoffs (CSR → Branch Officer → Operations Team) with no shared digital workflow, increasing turnaround time and risk of lost files.
- Debit card and internet banking activation are handled as separate, sequential manual requests instead of running in parallel with account creation.
- No visibility for the customer into application status between submission and account activation.
- Physical document storage creates compliance and retrieval risk.

Business Rules

- Customer must be physically present with original documents for KYC — no remote or digital verification permitted.
- Application form must be complete before being forwarded to the Branch Officer.
- Only the Branch Officer can approve or reject a new account application.
- Debit card and internet banking requests cannot be initiated until the account is confirmed created in CBS.
- Customer must be 18 years or older to open an individual account.

Inputs and Outputs

Inputs: Original ID proof, original address proof, passport-size photograph, filled paper application form, nominee details

Outputs: Active bank account with account number, physical welcome letter, debit card (dispatched separately), internet banking credentials, SMS/notification confirming activation

Opportunities for Improvement

- Replace physical KYC with digital eKYC (Aadhaar/PAN-based verification with video KYC).
- Eliminate paper forms in favor of a digital application captured directly into the system.
- Enable parallel processing of debit card and internet banking requests instead of sequential handling.
- Provide real-time application status visibility to the customer.
- Reduce manual handoffs between CSR, Branch Officer, and Operations Team through a shared digital workflow queue.

BPMN-Style Process Flow Description

Pool: ABC Bank Ltd. National Bank — Branch Account Opening (As-Is)

Lane: Customer

- Start Event: Customer arrives at branch
- Task: Take token, wait in queue
- Task: Provide original documents for KYC
- Task: Fill paper application form
- Task: Attach photocopies and photograph

Lane: Customer Service Representative (CSR)

- Task: Perform manual KYC verification
- Task: Issue application form
- Task: Review form completeness
- Gateway (Exclusive): Form complete?
 - No → Task: Return form to customer (loop to Customer lane)
 - Yes → Task: Log application, forward to Branch Officer

Lane: Branch Officer

- Task: Cross-check documents against application
- Gateway (Exclusive): Application approved?
 - No → Task: Notify customer of rejection → End Event: Application rejected
 - Yes → Task: Forward file to Operations Team

Lane: Operations Team

- Task: Enter customer data into CBS
- Task: Request debit card issuance
- Task: Request internet banking activation

Lane: Core Banking System (CBS)

- Task: Generate account number and create record
- Task: Provision internet banking access
- End Event: Notification sent to customer, account active

Pain Points Identified by the Business Analyst

1. Physical presence requirement excludes customers who cannot easily visit a branch, contributing to the onboarding delays flagged in the Business Problem.
2. Sequential, non-integrated handoffs between CSR, Branch Officer, and Operations Team introduce delay and risk of file loss with no digital audit trail.
3. Manual KYC and data entry are error-prone and slow compared to automated verification against government databases.
4. Debit card and internet banking are treated as afterthoughts rather than part of a single onboarding transaction, extending time-to-first-use well beyond account creation.
5. Customers have no self-service visibility into where their application stands, driving avoidable branch and call-centre follow-ups.

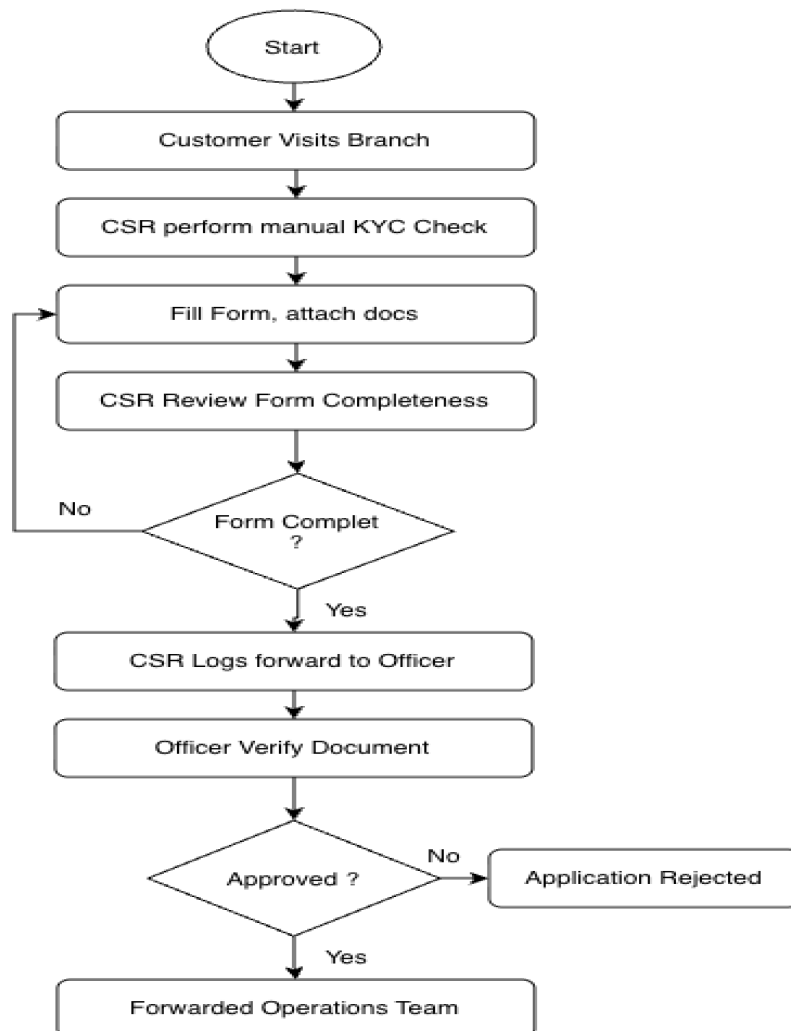
Recommendations for the To-Be Process

- Replace manual KYC with digital eKYC (Aadhaar/PAN validation + video KYC) to remove the branch dependency entirely for standard cases.
- Digitize the application form so customer-entered data flows directly into the platform, removing manual re-keying by Operations Team.
- Introduce a risk-based approval model: auto-approve standard applications, route only flagged/mismatched cases to manual compliance review.

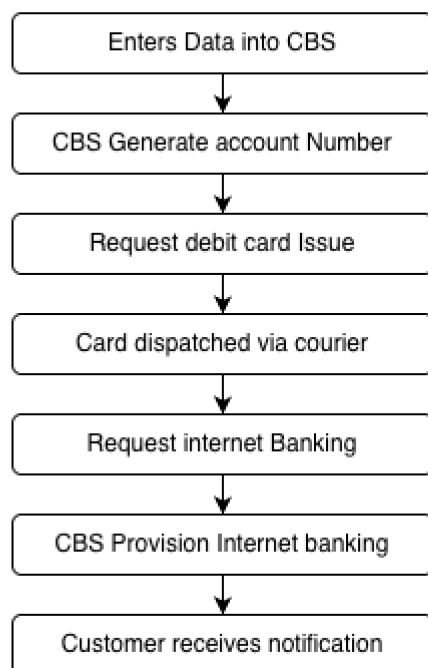
- Trigger debit card issuance and internet banking activation automatically and in parallel upon account creation, rather than as separate sequential requests.
- Provide customers a real-time application status tracker to eliminate status-check calls and visits.

As-Is Process Flow — Part 1: Branch Application & Approval

Part 1: Branch Application & Approval



As-Is Process Flow — Part 2: Backend Processing & Activation



Process Name

Digital Customer Onboarding — Self-Service Account Opening (To-Be)

Business Objective

To enable a retail customer to open a fully functional bank account — including internet banking and a debit card — entirely through the web or mobile app, without visiting a branch, using automated eKYC verification and system-to-system integration to reduce onboarding time from 2-3 days to same-day activation.

Actors

- **Customer** — individual opening a new account via web or mobile app
- **Digital Banking Platform** — the customer-facing application orchestrating the journey
- **eKYC Service** — third-party/government-integrated identity verification service (Aadhaar/PAN, video KYC)
- **Core Banking System (CBS)** — system of record for account creation
- **Notification Service** — sends SMS/email/in-app alerts

- **Bank Administrator** — reviews only flagged/exception cases

Step-by-Step Future Process (18 Steps)

1. Customer downloads the app or visits the web platform and selects "Open New Account."
2. Customer enters mobile number and email address.
3. Platform sends OTP to mobile number and email simultaneously.
4. Customer enters both OTPs to verify contact details.
5. **Decision Point:** Are both OTPs valid? If invalid, customer is prompted to re-enter or request a resend (loop to Step 3).
6. Customer enters PAN and Aadhaar number.
7. Platform calls the eKYC Service to validate identity against UIDAI/NSDL in real time.
8. Customer uploads a photo ID and completes a live video KYC session within the app.
9. eKYC Service cross-checks the video KYC capture against the submitted ID document (face match, liveness check).
10. **Decision Point:** Does eKYC pass validation? If a mismatch or low-confidence match is detected, the application is routed to the Bank Administrator for manual review; customer is notified their application is under review (exception path).
11. If eKYC passes automatically, the platform proceeds to digital account opening — customer selects account type and reviews terms.
12. Customer digitally accepts terms and conditions and submits the application.
13. Platform sends the validated application data to the Core Banking System via API.
14. CBS automatically creates the account and generates an account number in real time.
15. Platform triggers instant Internet Banking activation upon account creation confirmation from CBS.
16. Customer sets up login credentials and registers for Multi-Factor Authentication (OTP and/or biometric).
17. Platform submits a parallel automated request for debit card issuance (physical + optional instant virtual card).
18. Notification Service sends a real-time welcome email and SMS confirming the account number, internet banking activation, and card dispatch status.

Decision Points

- OTP validity check for mobile and email (Step 5)
- eKYC pass/fail validation, determining automated approval vs. manual review routing (Step 10)

Automated System Activities

- OTP generation and validation
- Real-time PAN/Aadhaar validation against government databases
- Face match and liveness detection during video KYC
- Automatic account number generation in CBS
- Automatic Internet Banking provisioning
- Automatic debit card issuance request
- Automated real-time notifications at each key milestone

System Integrations

- Digital Banking Platform ↔ eKYC Service (identity verification API)
- Digital Banking Platform ↔ Core Banking System (account creation API)
- Digital Banking Platform ↔ Notification Service (SMS/email/push gateway)
- Core Banking System ↔ Card Management System (debit card issuance trigger)
- Digital Banking Platform ↔ MFA Provider (OTP/biometric authentication service)

Business Rules

- Mobile number and email must each be verified via OTP before proceeding to eKYC.
- Customer must be 18 years or older; the system shall auto-reject underage applicants at the PAN/Aadhaar validation step.
- Video KYC must achieve a minimum face-match confidence threshold; below this, the application routes to manual review, never auto-rejects.
- Internet Banking shall only activate after CBS confirms successful account creation.
- MFA registration is mandatory before first login is permitted.
- Debit card issuance request shall trigger automatically and in parallel with Internet Banking activation, not sequentially.

Inputs and Outputs

Inputs: Mobile number, email address, PAN, Aadhaar number, live video KYC capture, digitally accepted terms and conditions

Outputs: Active bank account with account number, activated Internet Banking credentials, MFA registration, debit card dispatch request confirmation, real-time welcome email/SMS

Exception Scenarios

- **OTP not received:** Customer can request a resend after a cooldown period; repeated failures direct the customer to alternate verification (registered branch or support).
- **eKYC face-match failure:** Application routed to Bank Administrator queue for manual review; customer notified of delay, not rejection.
- **PAN/Aadhaar validation service downtime:** Application is held in a pending state with customer notification; automatic retry attempted before manual fallback.
- **CBS account creation failure:** Platform retries automatically; if unresolved, the case is escalated to the Bank Administrator with the customer notified of the delay.
- **Underage applicant:** System auto-rejects at PAN/Aadhaar validation with a clear on-screen message; no manual review needed

BPMN-Style Process Flow Description

Pool: ABC Bank Ltd. — Digital Customer Onboarding (To-Be)

Lane: Customer

- Start Event: Customer opens app/web platform
- Task: Enter mobile number and email
- Task: Enter OTPs received
- Task: Enter PAN and Aadhaar
- Task: Complete video KYC
- Task: Select account type, accept terms
- Task: Set up MFA credentials

Lane: Digital Banking Platform

- Task: Send OTP to mobile and email
- Gateway (Exclusive): OTPs valid?

- No → Task: Prompt resend (loop to Customer lane)
- Yes → Task: Call eKYC Service
- Task: Submit application data to CBS
- Task: Trigger Internet Banking activation
- Task: Trigger parallel debit card request

Lane: eKYC Service

- Task: Validate PAN/Aadhaar against government database
- Task: Perform face-match and liveness check on video KYC
- Gateway (Exclusive): eKYC passed?
 - No → Task: Route to Bank Administrator (manual review)
 - Yes → Task: Return validation success to platform

Lane: Core Banking System

- Task: Create account record
- Task: Generate account number
- Task: Confirm creation to platform

Lane: Notification Service

- Task: Send OTP messages
- Task: Send welcome email/SMS with account and activation details
- End Event: Customer onboarded and notified

Lane: Bank Administrator

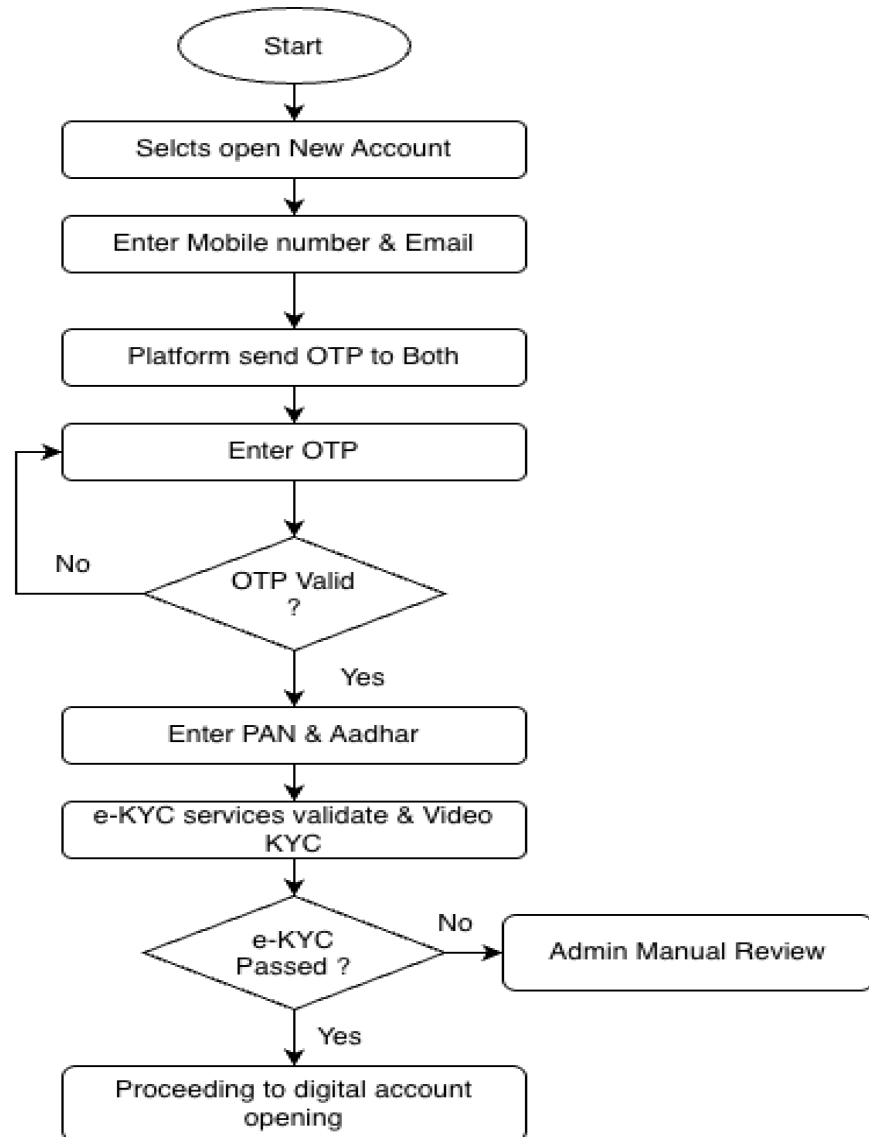
- Task: Review flagged eKYC exceptions
- Gateway (Exclusive): Manual approval granted?
 - No → End Event: Application rejected, customer notified
 - Yes → Task: Resume automated flow (rejoin main path)

Comparison: As-Is vs. To-Be

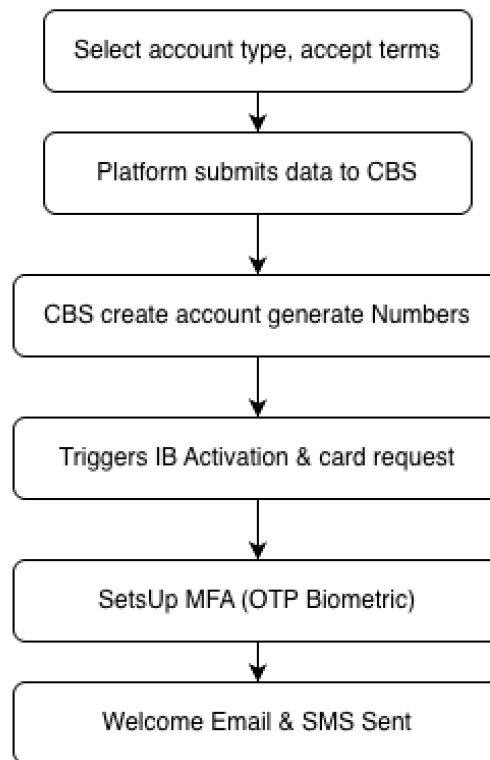
Aspect	As-Is	To-Be
Channel	Branch only	Web and mobile app
Identity verification	Manual, physical document check	Automated eKYC with video verification
Application format	Paper form	Fully digital

Turnaround time	2-3 working days	Same-day, often minutes
Human involvement	CSR, Branch Officer, Operations Team at every step	System-automated; human involved only for flagged exceptions
Card & Internet Banking activation	Sequential, separately requested	Parallel, automatic
Authentication	Single-factor login	Mandatory MFA
Status visibility	None until completion	Real-time in-app and notification updates one
Compliance/audit trail	Manual register, physical files	Fully digital, system-logged

To-Be Process Flow — Part 1: Verification & eKYC



To-Be Process Flow — Part 2: Account Creation & Activation



14. Challenge & Solutions

Challenge 1: Balancing security friction with usability in MFA

Introducing mandatory MFA on every login risked frustrating customers and increasing drop-off, while the Fraud & Risk team pushed for the strictest possible controls.

Solution: Ran a joint workshop with Fraud & Risk and IT to agree on a one-extra-step limit (NFR-07), offering both OTP and biometric as interchangeable second factors so customers could choose the faster option without weakening security.

Challenge 2: Conflicting requirements between Compliance and Customer Experience

Compliance wanted extensive manual document review for every eKYC submission, which would have defeated the goal of same-day onboarding.

Solution: Proposed a risk-based validation model — automated checks for standard cases, with manual review triggered only when the system detects a name/DOB mismatch or other red flag (FR-02). This preserved compliance oversight without slowing down the majority of applications.

Challenge 3: Legacy core banking system (CBS) integration constraints

Early technical workshops revealed the existing CBS did not expose real-time APIs for all required data (e.g., FD interest calculations), risking delays to the Account Summary and Fixed Deposit modules.

Solution: Worked with the IT team to sequence delivery — modules dependent on available APIs were prioritized first, while a phased API development plan was agreed with the CBS vendor for the remaining data points, and this dependency was documented in the Assumptions & Constraints section.

Challenge 4: Scope creep from stakeholder requests

Multiple stakeholder groups (Relationship Managers, Loan Processing Team) requested additional features during workshops that fell outside the board-approved roadmap, such as investment product cross-selling.

Solution: Introduced a formal change request process tied to the Scope document, requiring BA sign-off before any new feature was added — keeping the 12-module scope intact while logging out-of-scope requests for a potential future phase.

Challenge 5: Fragmented input from a wide stakeholder base

With 10 distinct stakeholder groups involved (per the Stakeholder Analysis), individual interviews and workshops surfaced overlapping or contradictory requirements — particularly between Customer Support and Fraud & Risk on how much account information a support executive should be able to view.

Solution: Consolidated conflicting inputs into a single requirements review workshop where both teams jointly defined role-based access boundaries, later reflected in the Admin Portal's role-based access control requirement (FR-16).

Challenge 6: Validating requirements without a working prototype

Business stakeholders struggled to visualize how digital onboarding or loan tracking would actually function from written requirements alone.

Solution: Used low-fidelity wireframe walkthroughs during the elicitation phase (per the Requirement Gathering Approach) to validate assumptions before finalizing the FR/NFR document, reducing rework during later UAT.

Challenge 7: Migrating legacy customer data without service disruption

Existing customer records needed to move from three legacy systems into the unified platform, with risk of data loss or mismatched account histories.

Solution: Recommended a parallel-run period with data reconciliation checks before final cutover, documented as a mitigation under the Risks register, ensuring customers saw no discrepancy in balances or transaction history at go-live

15. Business Impact

1. Customer Impact

- **Faster onboarding:** Account activation time drops from 2-3 working days to same-day, directly addressing the customer acquisition losses to digital-first competitors identified in the Business Problem.
- **Reduced friction on high-value products:** Fixed deposits and personal loans move from branch-dependent to fully self-service, removing the in-person step that was causing loan application abandonment.
- **Greater transparency:** Real-time notifications and application tracking (Notifications, Personal Loan modules) reduce the "black box" experience customers previously had with branch-based processes.

2. Operational Impact

- **Lower call-centre load:** Self-service access to balances, statements, and FD status (Dashboard, Account Summary) is expected to reduce informational call volume, freeing support staff for higher-value escalations.
- **Reduced branch dependency:** Relationship managers and branch staff are no longer required for routine onboarding or FD/loan initiation, allowing branch capacity to be redirected toward advisory and complex servicing.
- **Faster loan processing:** A consolidated digital queue (Admin Portal, Personal Loan Application) replaces manual document coordination, shortening the loan processing team's turnaround time.

3. Financial Impact

- **Reduced onboarding and servicing cost per customer:** Digital eKYC and self-service tools lower the operational cost of acquiring and servicing each customer compared to branch-based processes.
- **Increased product adoption:** Removing branch dependency for FDs and personal loans is expected to increase digital origination volume for these products, directly supporting fee and interest income growth.
- **Lower fraud-related losses:** Mandatory MFA and real-time fraud flagging (Login with MFA, Admin Portal fraud dashboard) reduce exposure to account takeover losses previously tied to single-factor authentication.

4. Risk & Compliance Impact

- **Regulatory alignment:** Enforced MFA and structured eKYC bring the bank into alignment with RBI's Master Direction on Digital Payment Security Controls, closing a gap the risk team had already flagged.
- **Improved auditability:** Centralized logging of onboarding, transactions, and MFA activity (NFR-08, Admin Portal) gives compliance teams faster, more reliable audit reporting than manually compiled branch records.

5. Competitive Impact

- **Retention against digital-first competitors:** Matching neobank-level onboarding speed and self-service capability directly addresses the attrition trend cited in the Business Problem, positioning ABC Bank Ltd. to compete on digital experience rather than branch footprint alone.

16. Business Analysis Techniques & Deliverables

Technique	Purpose	Deliverable Produced
Stakeholder Analysis	Identify and categorize stakeholders by interest and influence	Stakeholder Analysis document
Document Analysis	Review existing net banking specs, RBI guidelines, complaint logs	Business Problem, Assumptions & Constraints
Interviews	One-on-one elicitation with Loan Processing, Compliance, IT	Functional/Non-Functional Requirements

Workshops	Joint sessions with Fraud & Risk and IT on MFA and security	Functional/Non-Functional Requirements, Risks
Process Observation (Shadowing)	Observe support executives handling live calls	Requirement Gathering Approach, Notifications / Customer Support requirements
Surveys	Gauge customer digital adoption barriers and feature priorities	Requirement Gathering Approach, Business Objectives
Prototype Walkthroughs	Validate assumptions using low-fidelity wireframes before finalizing FRs	Requirement Gathering Approach, Challenges & Solutions
MoSCoW / Priority Ranking	Rank requirements by priority (High/Medium)	User Stories, Functional Requirements
User Story Mapping	Break down modules into role-based stories with acceptance criteria	User Stories (24 stories, 11-field format)
Given-When-Then (Gherkin)	Define testable conditions for each user story	Acceptance Criteria
SWOT-style Risk Identification	Identify risks with impact, likelihood, and mitigation	Risk Register
As-Is/To-Be Process Mapping	Compare current branch-based process against proposed digital process (Branch-based banking process) to the proposed digital process	As-Is/To-Be Onboarding Process Flow Diagram
Requirement Traceability	Trace user stories to requirements, objectives, and stakeholders	Requirement Traceability Matrix
UAT Scenario Design	Define test scenarios validating acceptance criteria	UAT Scenarios, UAT Summary Report
Root Cause / Gap Analysis	Identify underlying causes behind onboarding delays, call volume, fraud exposure	Business Problem, Challenges & Solutions
Impact Analysis	Translate objectives into measurable customer, operational, financial, and risk outcomes	Business Impact document

Tools: Jira, Confluence, Figma, Excel, SQL, Power BI, MS Visio / Draw.io.

17. Key Learning

Working through this case study reinforced a few things that don't always come across in requirement templates alone.

The biggest one was how much time gets saved by validating requirements before they're finalized rather than after. Walking stakeholders through low-fidelity wireframes early — before the FR/NFR document was locked — caught issues like MFA friction that would have been far more expensive to fix during UAT. It's tempting to treat prototyping as a design-phase activity, but here it functioned as a requirement validation tool, and that shift in mindset was probably the single most useful lesson from this project.

Conflicting stakeholder priorities also turned out to be more the norm than the exception. Compliance wanting exhaustive manual review and the business wanting same-day onboarding weren't really in conflict once reframed as a risk-based validation model — the tension wasn't about disagreement on the goal, it was about each team defaulting to their own worst-case assumption. A good chunk of BA work here wasn't writing requirements at all; it was translating one team's concern into language the other team could act on.

Traceability mattered more than expected. Building the Requirement Traceability Matrix felt like overhead early on, but it became the fastest way to answer "why do we need this feature" during later reviews — every requirement had a stakeholder and a business objective attached, so nothing needed re-justifying from scratch.

Technical constraints surfaced late are a real risk, not just a theoretical one. The CBS integration gaps only became clear during technical workshops, well after initial scope was drafted. In hindsight, involving IT earlier — even informally — during the Business Problem and Scope stages would have avoided some rework in sequencing the module rollout.

Finally, writing acceptance criteria in Given-When-Then format before development, not after, made UAT genuinely fast. UAT validation was performed against 24 defined scenarios. The identified defects were documented, tracked and resolved before final UAT sign-off. which wasn't luck — it was a direct result of the acceptance criteria already being unambiguous and testable from the start, rather than being reverse-engineered from working software.

If I were to carry one habit forward from this project, it would be this: treat prototyping and traceability as elicitation tools, not just documentation artifacts — they surface problems while they're still cheap to fix.

18. Conclusion

The Digital Retail Banking Platform (DRBP) existing digital channels were built as isolated, transaction-only tools rather than a complete banking experience — a gap that was costing the bank customer acquisition, straining branch and support capacity, and exposing it to regulatory and fraud risk. The proposed Digital Retail Banking Platform addresses this directly by unifying onboarding, day-to-day banking, lending, and customer support into a single, secure, self-service experience across all 12 required modules.

Through structured requirement gathering, stakeholder engagement across 10 distinct groups, and a full traceability chain from business objectives through user stories to acceptance criteria, this case study demonstrates how a fragmented digital banking journey can be redesigned around measurable outcomes: same-day account activation, reduced call-centre dependency, faster loan turnaround, and closed authentication security gaps.

UAT validation confirmed all 24 scenarios passed against their defined acceptance criteria, and the As-Is/To-Be process comparisons across every module show a consistent shift — from manual, branch-dependent steps to automated, real-time digital journeys. Combined with the Business Impact analysis, this positions the platform not just as a technology upgrade, but as a direct response to a clearly defined business problem, with success measured against concrete targets rather than general digital transformation goals.

This case study reflects an end-to-end Business Analyst workflow — from problem identification through requirement elicitation, documentation, traceability, process design, and UAT sign-off — suitable as a standalone portfolio artifact.